TIME SERIES FOUNDATION MODELS

TSFMs for CPG forecasting: lessons from Unilever H3L Labs

H3L forecasting team

Unilever H3L Labs · Applied ML

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Who we are and why we forecast

Unilever

One of the world's largest consumer-goods companies — brands used by billions of people every day, built on €16B+ of materials bought each year.

Horizon 3 AI Labs

Unilever's AI innovation lab — named for the third horizon: exploring frontier AI research and turning it into working products for the business.

Competitive Buying

Buying smarter in a competitive, complex supply chain — targeting +1% material-spend efficiency, worth ≈ €160M a year.

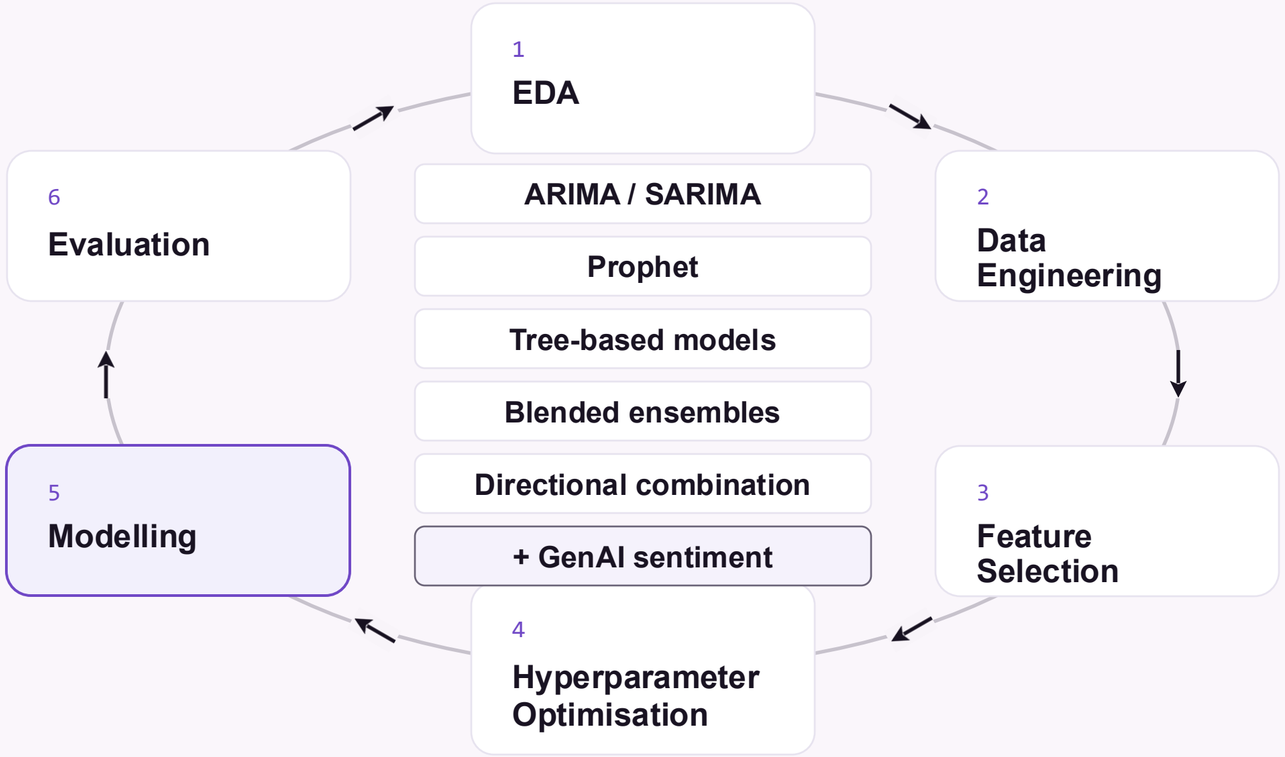
Our running examples in this talk:

- Fatty Alcohol (FAL)
- Crude Palm Oil (CPO)
- Palm Kernel Oil (PKO)

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How we forecast today

A competitive buying environment and a complex supply chain — we optimise how we buy.



THE TOOLKIT KEPT EVOLVING

Each generation added capability — nonlinearity, stability, direction, and market context.

ASEP — our GenAI pipeline reads market reports & news and turns them into sentiment covariates.

But every one of these is still a bespoke model, fit to a single series

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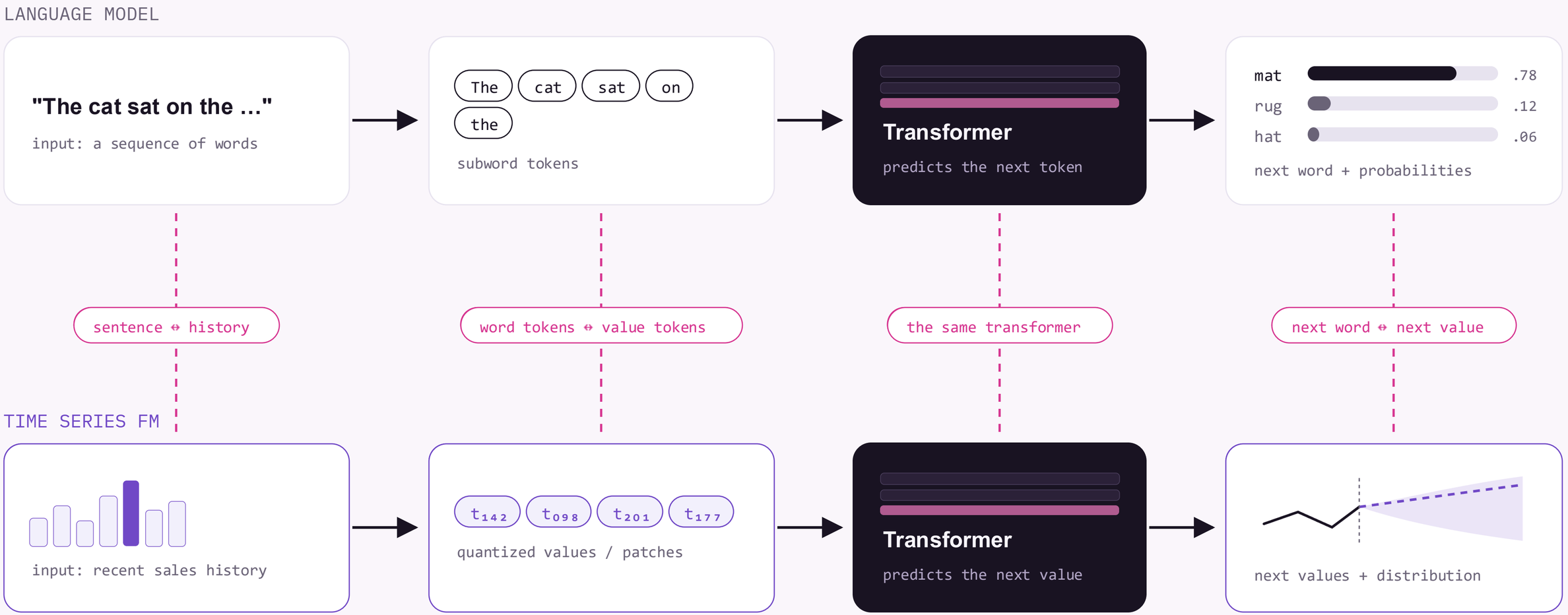
Why look beyond classical ML models

To answer this question, let's first understand what TSFMs actually are?

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TSFMs apply the LLM analogy to numeric sequences



Same recipe — the vocabulary is numeric. Which is why **tokenization design** is where architectures diverge.

The design space: every TSFM is one path through five steps

1 INPUT + SCALING



Raw series

scaled per window

2 · TOKENIZATION

t_{142} t_{098} t_{201}

Quantize to a discrete vocab

Chronos



Real-valued patches

TimesFM · Moirai · Toto



Lag-feature vectors

Lag-Llama

3 · EMBED + POSITION

Embed + positional encoding

4 · TRANSFORMER — one of three

Encoder-only

bidirectional attention

Moirai

Decoder-only

causal attention

TimesFM · Lag-Llama · Toto

Encoder–decoder

encode once, cross-attend

Chronos (T5)

5 · FORECAST HEAD



Sampled paths → quantiles

Chronos



Student-t parameters

Lag-Llama



Mixture of distributions

Moirai · Toto

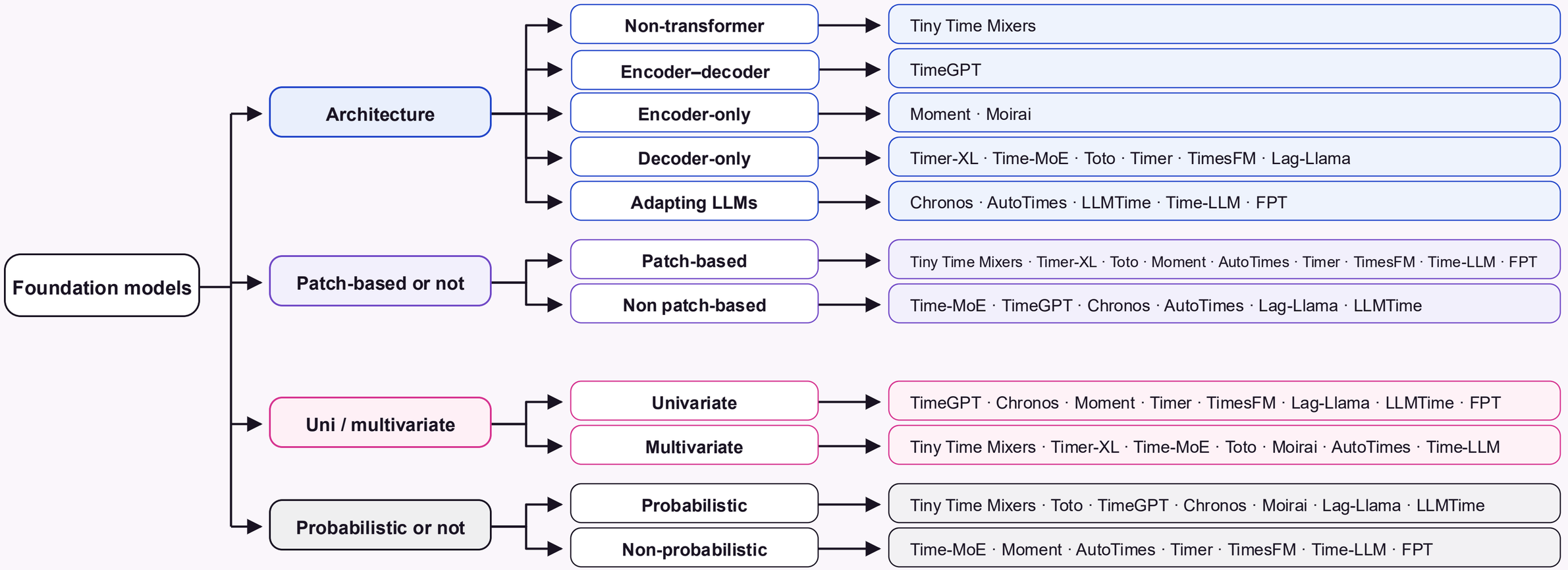


Point forecast — the exception

TimesFM

AUTOREGRESSIVE LOOP — forecasts re-enter the context

Classification of TSFMs



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TSFM changes the following

A single model pretrained on billions of time points across many domains — forecasts a brand-new series zero- or few-shot, no per-series training, with native probabilistic output and support for external covariates.

OUR CEILING · A BESPOKE MODEL FOR EVERY SERIES	WHAT A TSFM CHANGES · ONE FOUNDATION MODEL
A separate model for every commodity	One pretrained model forecasts any series — no more per-series builds
Weeks of training & tuning per model	Zero-shot: forecasts in minutes — no training loop at all
Too little history to forecast new series	Has already learned from thousands of similar series — useful from day one
Point forecast, no sense of risk	Probabilistic forecast quantifies uncertainty

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Forecasting Approaches Undertaken

Two approaches are being tested — zero-shot forecasting , fine-tuning .

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Two approaches: zero-shot , fine-tuning



Zero-shot forecasting COMPLETED

The frozen pre-trained model forecasts an unseen series purely from the recent history in its context window.

+ scales

+ fast

– not domain specific

– accuracy ≈ classical

Fine-tuning IN PROGRESS

Continue training the pre-trained weights on years of our own history — three options commonly used.

Full fine-tuning — all weights	compute-heavy · catastrophic-forgetting risk
OUR PICK LoRA — low-rank adapters (PEFT)	core frozen · ~1% of params trained
Multi-scale / TS-specific	frequency-aware adapters — experimental

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Zero Shot Forecasting Experiment

Univariate and Multivariate zero shot forecasting

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A few terms before the deep dive

`Context length`	How many past months we show the model before it forecasts.
`Cutoff`	The date we stand on. The model sees only before it; we test on what came after.
`Horizon`	How many months ahead we ask the model to predict.
`MAPE`	Our error metric— how far the forecast is from the real value, in percent. **Lower is better.**
`Release date`	The day a model was published. Earlier data may already be inside its training.

Univariate results

Best variant per family · mean MAPE vs production baseline · lower is better

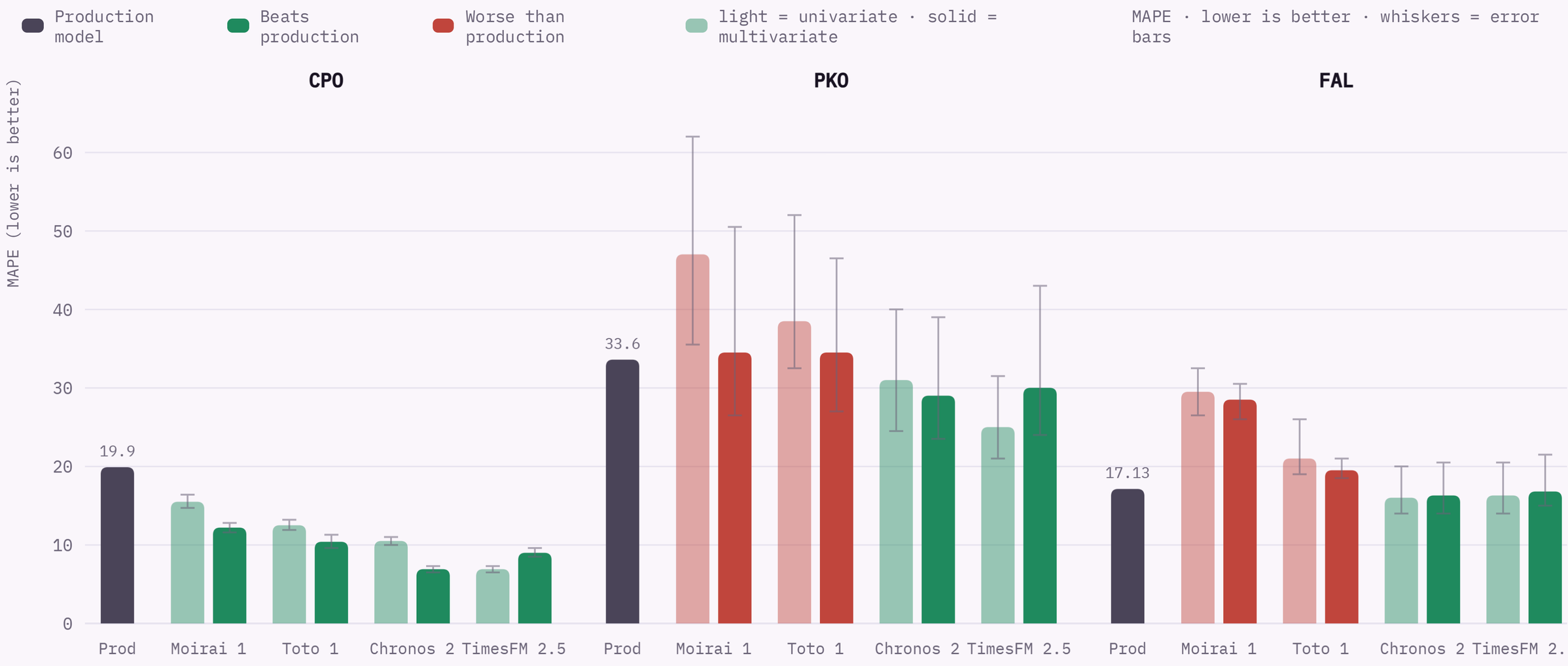
beat prod −15 +13 lost to prod

	CPO prod 19.9	PKO prod 33.6	FAL prod 17.13
Chronos	**6.71** −13.2 v: 1 · r: 3.8–14.5	**22.33** −11.3 v: bolt-small · r: 5.4–69.3	**10.30** −6.8 v: t5-mini · r: 5.2–19.3
Lag-Llama	**11.99** −7.9 v: lag_llama · r: 7.5–21.7	**34.99** +1.4 v: lag_llama · r: 9.6–119.2	**21.02** +3.9 v: lag_llama · r: 6.8–46.0
Moirai	**5.80** −14.1 v: 2.0-R-small · r: 4.3–10.3	**25.27** −8.3 v: 2.0-R-small · r: 4.7–67.4	**10.17** −7.0 v: 1.0-R-base · r: 6.0–17.8
MOMENT	**20.26** +0.4 v: 1-large · r: 3.0–48.6	**34.96** +1.4 v: 1-large · r: 11.1–83.1	**29.49** +12.4 v: 1-large · r: 9.6–63.8
Timer	**18.40** −1.5 v: base-84m · r: 4.6–60.8	**28.15** −5.5 v: base-84m · r: 11.7–63.4	**14.76** −2.4 v: base-84m · r: 6.7–26.1
TimesFM	**8.22** −11.7 v: 2.0-500m · r: 4.1–13.6	**21.25** −12.4 v: 1.0-200m · r: 3.4–49.4	**12.96** −4.2 v: 2.0-500m · r: 6.2–28.6
Toto	**5.57** −14.3 v: 1.0-base · r: 4.2–8.2	**23.81** −9.8 v: 2.0-1B · r: 2.8–86.0	**12.04** −5.1 v: 2.0-1B · r: 5.3–24.6
TTM	**19.75** −0.1 v: 2 · r: 3.4–77.0	**17.76** −15.8 v: 2 · r: 4.7–76.9	**15.92** −1.2 v: 2 · r: 3.8–58.4

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Multivariate Results: Does Multivariate Help?



Multivariate Results: Does Feature Selection help TSFM?

Univariate
Target price history only, no covariates — the bar every method must beat.

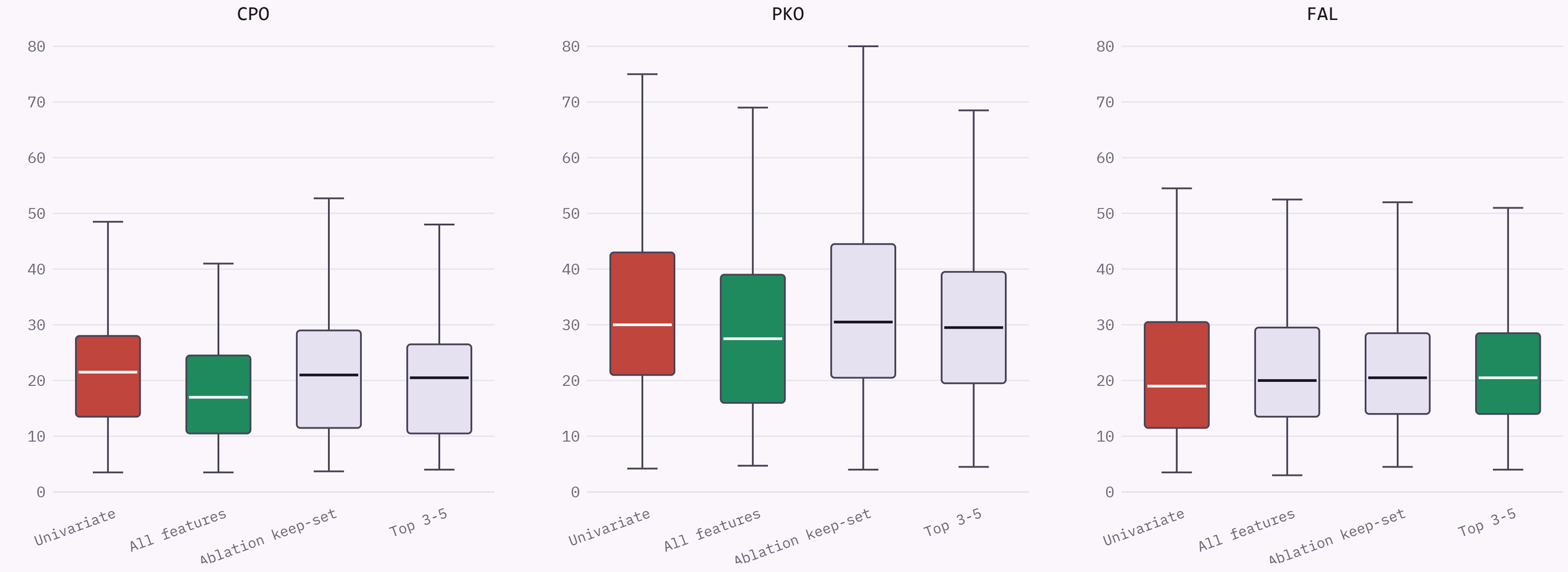
All features
Every available covariate column, switched on at once — no filtering.

Ablation keep-set
Covariates tested one at a time; only those that beat the baseline are kept.

Top 3–5
The 3–5 ablation covariates with the strongest lift — the leanest config.

Green box = lowest median MAPE in that panel

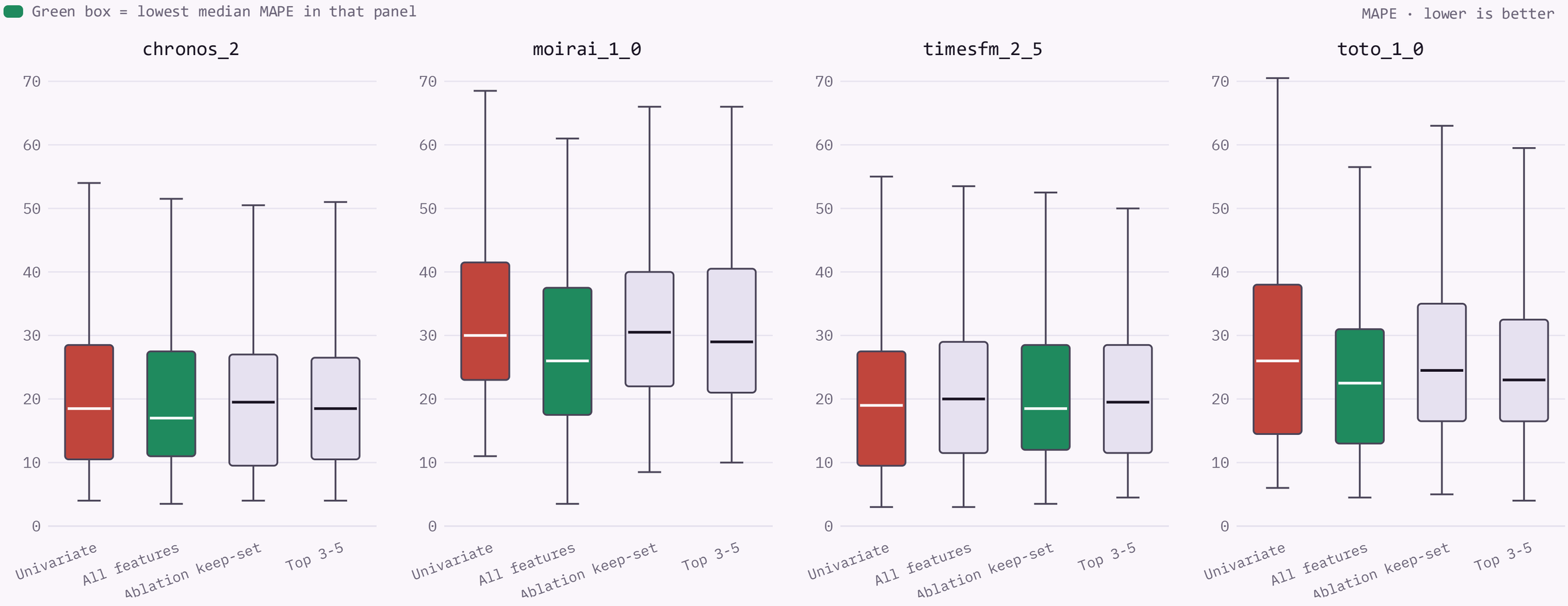
MAPE · lower is better



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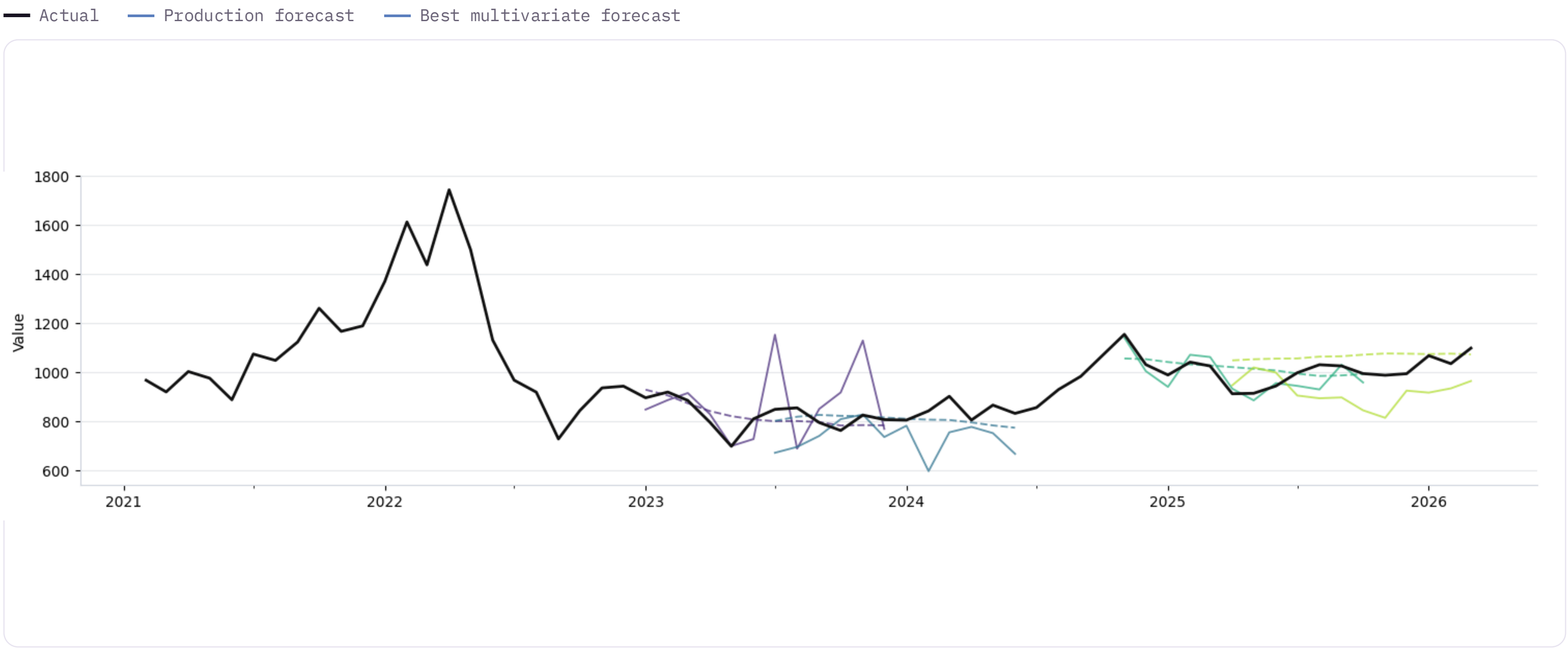
Multivariate Results: Do we see pattern among models for FS ?



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CPO forecast output

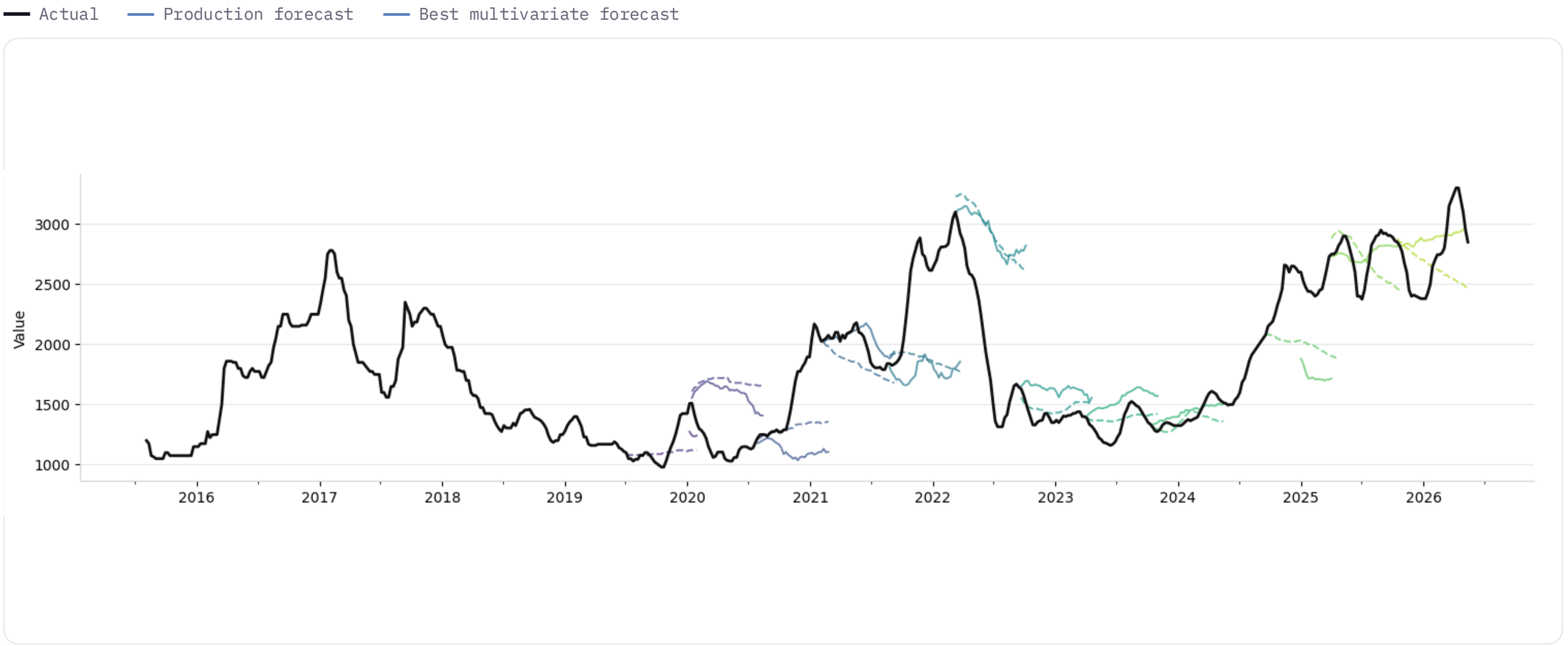
Actuals with paired production and best multivariate TSFM forecasts — 4 cutoffs, about 6 months apart.



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FAL forecast output

Actuals with paired production and best multivariate TSFM forecasts — 12 cutoffs, about 6 months apart.



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Learnings from Zero Shot Experiment

Cross-model comparison is confounded

Treat rankings as guidance, not conclusions for all domains

Covariates underdelivered on TSFMs

General-purpose foundation models are strong univariate zero-shot forecasters but weak at exploiting covariates and context ; Did not capture fluctuations well

Feature Selection doesn't improve results largely

Passing all clean data without selection gives better Accuracy

Context Length in a tuning parameter

Best length shifts per model and commodity — no rule

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Challenges and possible solutions

Fair Evaluation & Data Leakage

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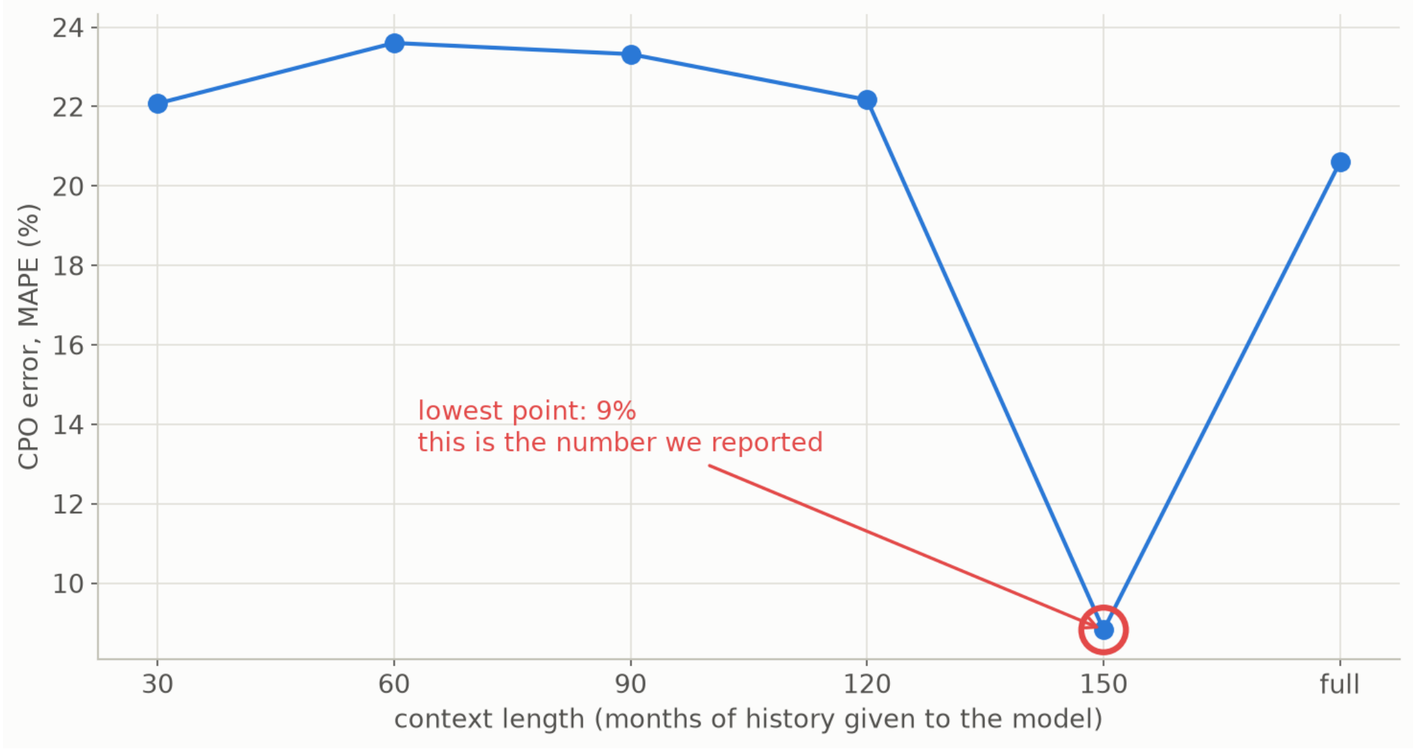
Evaluating a pretrained TSFM: 5 challenges

1	**Selection**	The same limited history must both choose the settings and measure the error.
2	**Coverage**	Long contexts only have eligible cutoffs at recent, calm dates, so settings are not automatically comparable.
3	**Sample size**	Few forecast dates per commodity, and neighbouring forecast windows overlap, so they are not independent.
4	**Regimes**	Calm and shock years produce very different errors, and one average mixes them.

+1 And one challenge sits outside our pipeline. The model was **pretrained on public data we cannot inspect**.

The lowest point of a tuning curve is an optimistic estimate

We tuned the model's input window and the best setting scored 9% error. This section asks how much of that is real.



01

Context length is how many past months the model reads before it forecasts.

02

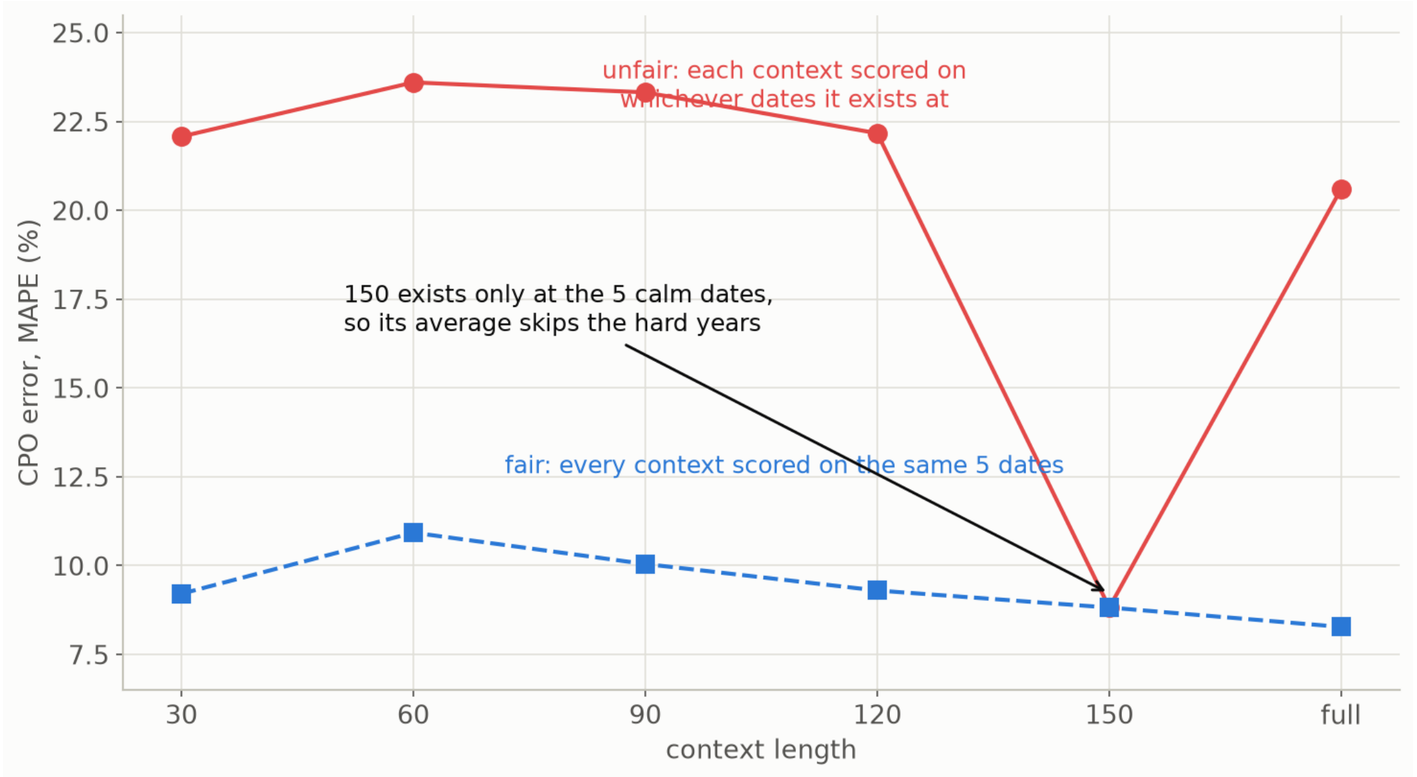
We tried six lengths and measured each one's error on past forecast dates.

03

We reported the best: 9%. Picking the minimum this way flatters the number.

Context length 150 was only evaluated on post-2023 cutoffs

CPO history starts in 2011, so long contexts were never scored on the shock years, and their average looked better than it was.



01

CPO data starts in 2011, so a 150-month context first exists around 2023.

02

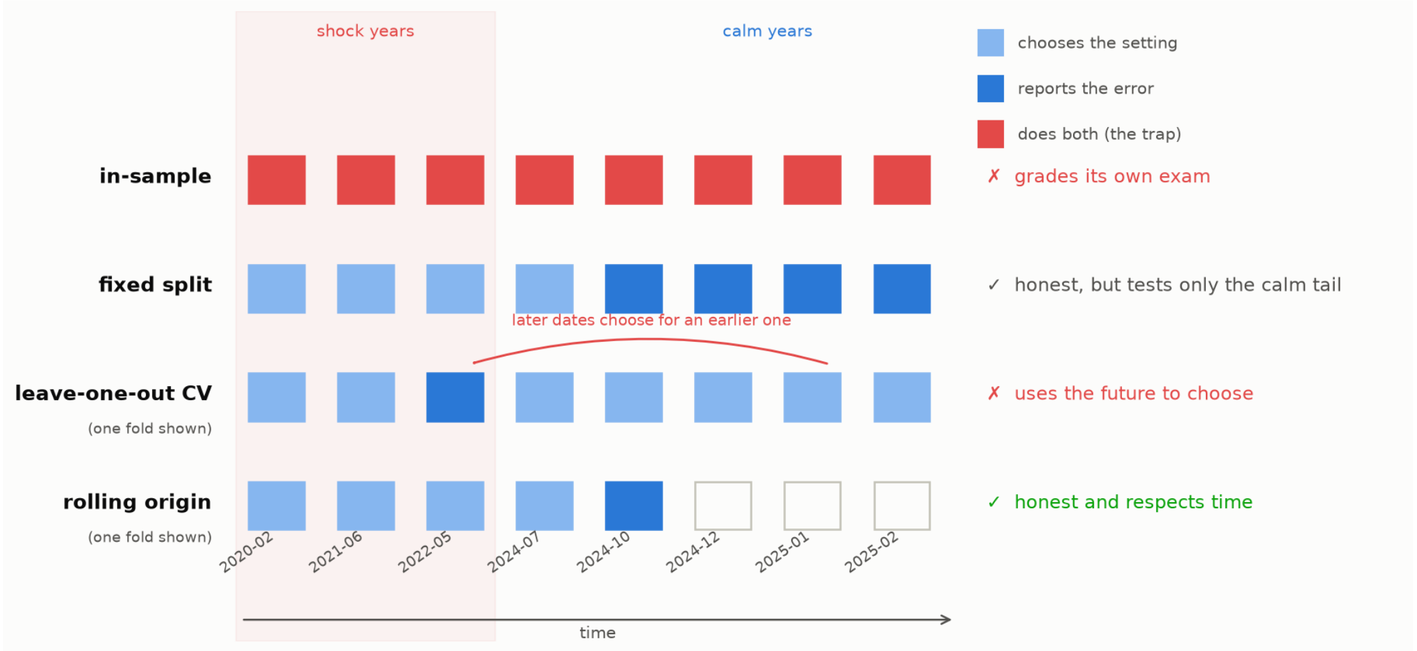
The price shocks of 2020 to 2022 happened before that.

03

Scored on the same dates, the dip at 150 disappears.

Selection and evaluation must use disjoint cutoffs

In-sample evaluation and leave-one-out CV both let test information into the choice.



01

Light blue dates choose the setting, dark blue dates report the error.

02

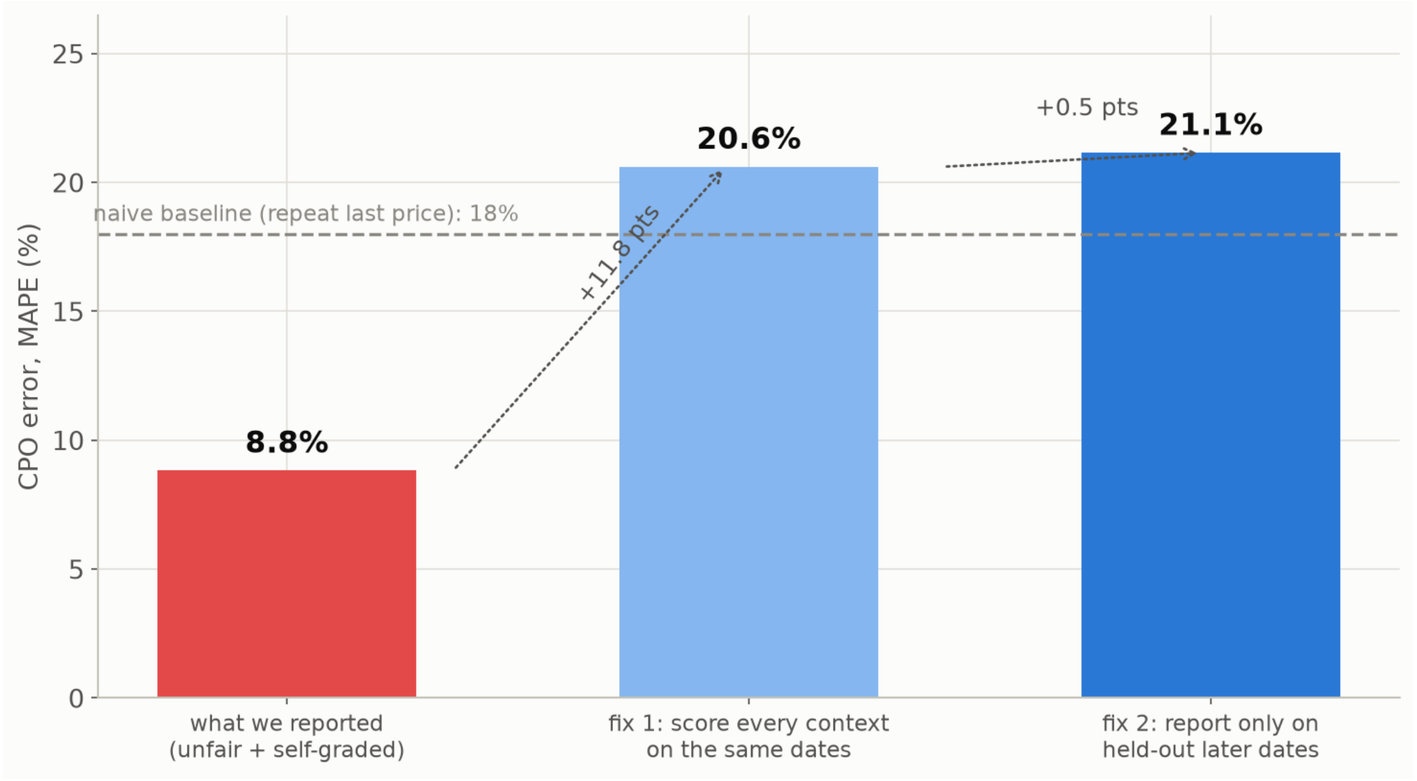
In-sample does both. Leave-one-out CV lets later dates choose for earlier ones.

03

Rolling origin and a gapped fixed split keep the two jobs separate.

After both corrections: 21% MAPE, against an 18% naive baseline

A corrected number still needs a reference point, so we compare against simply repeating the last known price.



01

Fix 1, same dates for every setting: 8.8 becomes 20.6.

02

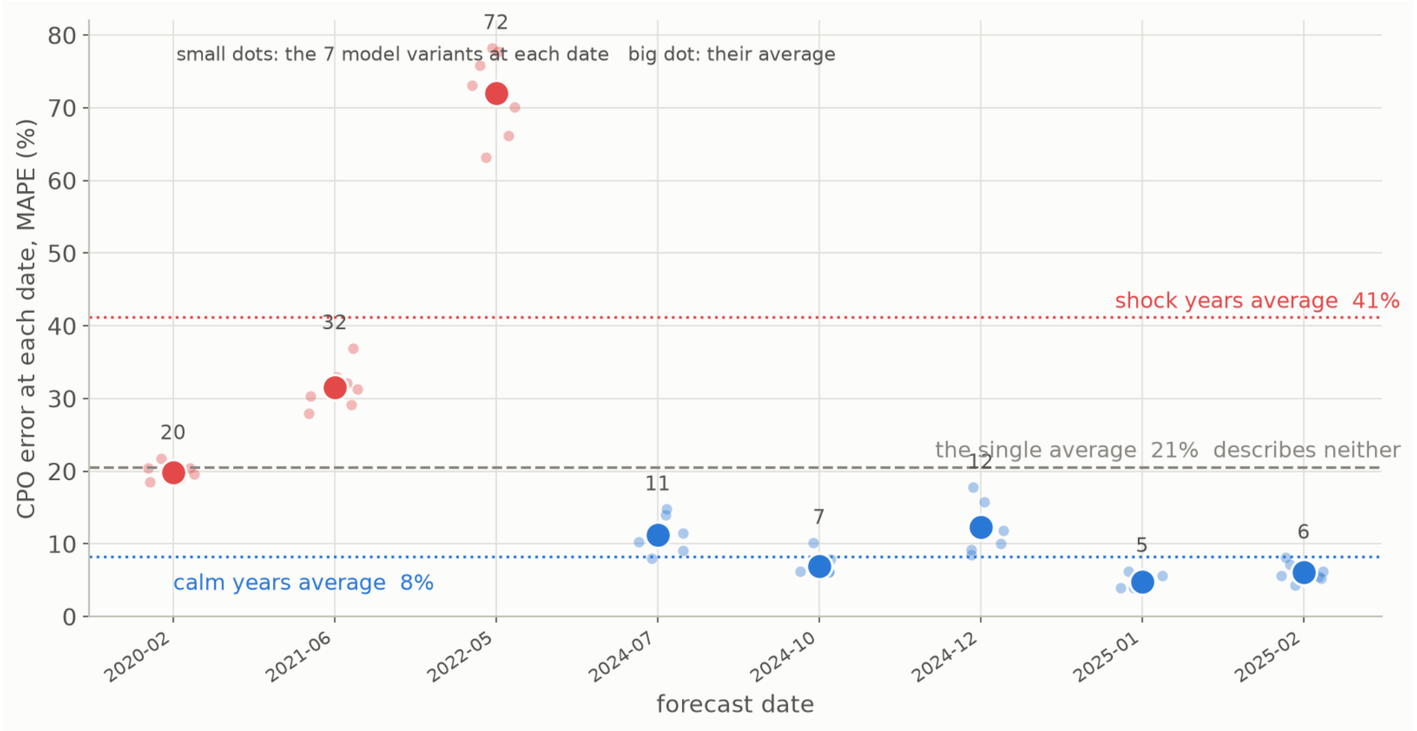
Fix 2, report only on held-out later dates: 21.1.

03

Repeating the last price scores 18. The gap to 21 is not decisive.

Error is regime-dependent: 8% in calm years, 41% in shocks

The 21% average mixes two very different periods and describes neither.



01

Each dot is the forecast error at one date.

02

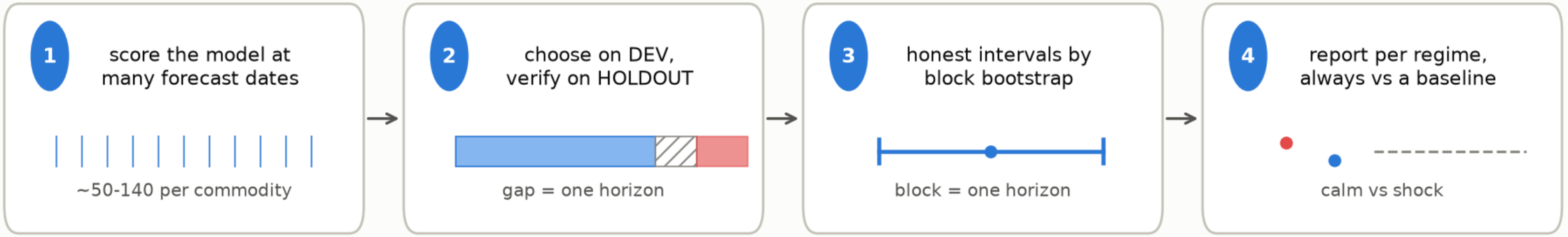
Calm years average 8 percent. Shock years average 41 percent.

03

For planning we quote the two regimes, never the blend.

Our evaluation protocol: dev/holdout split with a one-horizon gap

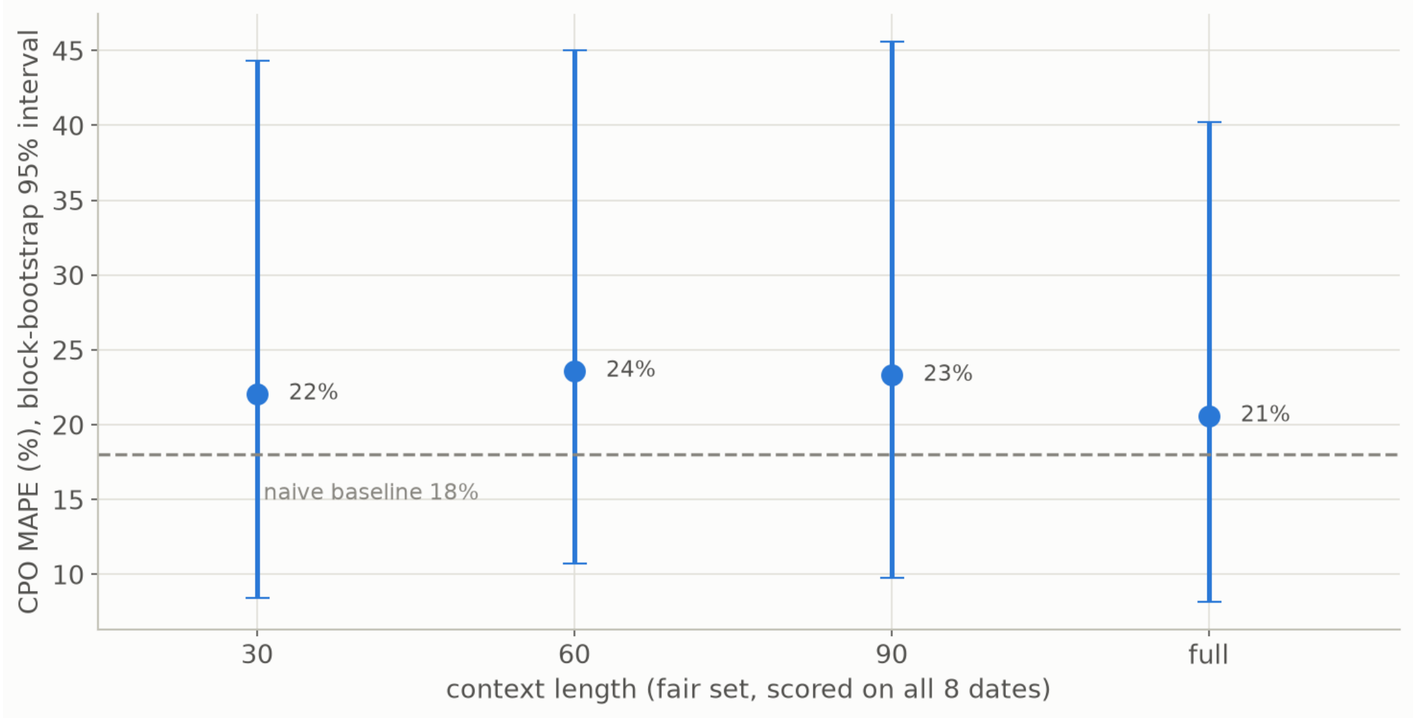
Fixing one number once is not enough, every future number needs the same discipline.



Score at many dates. Choose on early dates. Check on a held-out tail behind a one-horizon gap. Report per regime, always against a baseline.

Block bootstrap intervals: all context lengths are statistically tied

With about eight cutoffs per setting, a mean alone is unstable, and overlapping forecast windows are not independent.



01

Neighbouring cutoffs overlap, so we resample blocks of dates, not single dates.

02

The intervals say all context lengths are statistically tied.

03

None clearly beats the naive baseline.

Pretraining leakage: our test years may be in the training data

Commodity prices are public and we cannot inspect what the models were trained on.

Leakage we could cause in our pipeline

Temporal	Future information slips into preprocessing or tuning.
Covariate release	External data used before its real publication date.
Cross-series	A related series carries information from after the forecast date.

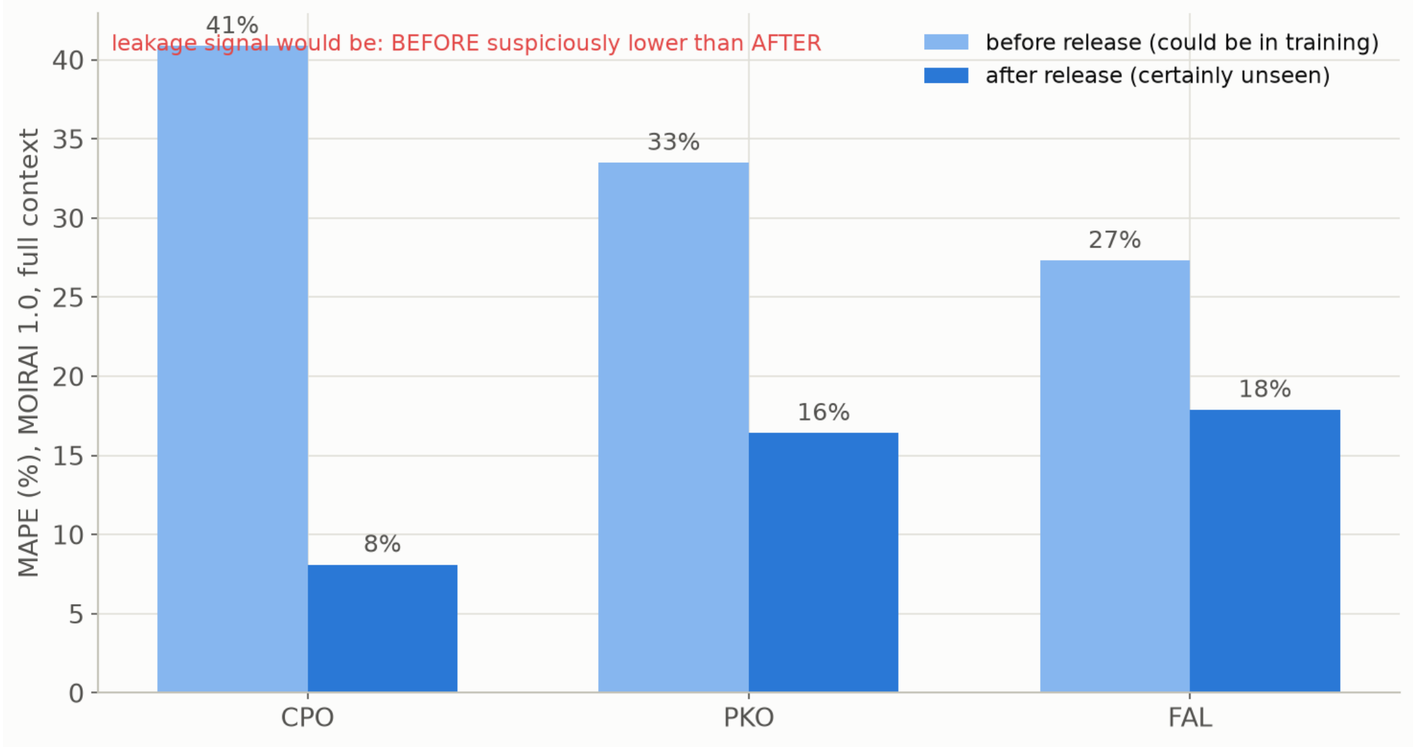
Leakage the model may carry from pretraining

Direct	Our exact series was in the training data.
Indirect	Something correlated was, like the same price from another vendor.

The **pretraining kind** matters most to us, because commodity prices are public.

Pre/post release check: no leakage signal, but low power

Only data published after the model release date is certainly unseen, and we have little of it.



01

Data published after the release date cannot be in the training set.

02

Error before release is higher, not suspiciously lower. No red flag.

03

Before-release years are also the shock years, so the test has low power.

SECTION 11 / 12

Our Next Steps

3 ideas we are working on

NEXT STEPS

Each open problem has a concrete next move

01

Detect leakage with influence functions

A memorized series leaves a fingerprint in the model's weights. We estimate what fraction of a benchmark the model memorized, without access to the training data.

status: under development

02

Fine-tune multivariate models

Zero-shot models read each series alone. We fine-tune multivariate models to learn the connections between price, supply, and sentiment series.

status: under development

03

Agentic forecasting, adapting Nexus (Google, 2026) to commodities

Our worst errors are in shock years, which are event-driven. We will put a TSFM inside the agent loop, feed it the sentiment signals already in our data platform, and test whether context reasoning cuts the shock-year error.

status: planned

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Q&A

Especially interested in your views on: fair evaluation of pre-trained forecasters · covariate injection · the case for a CPG-specific FM.

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